Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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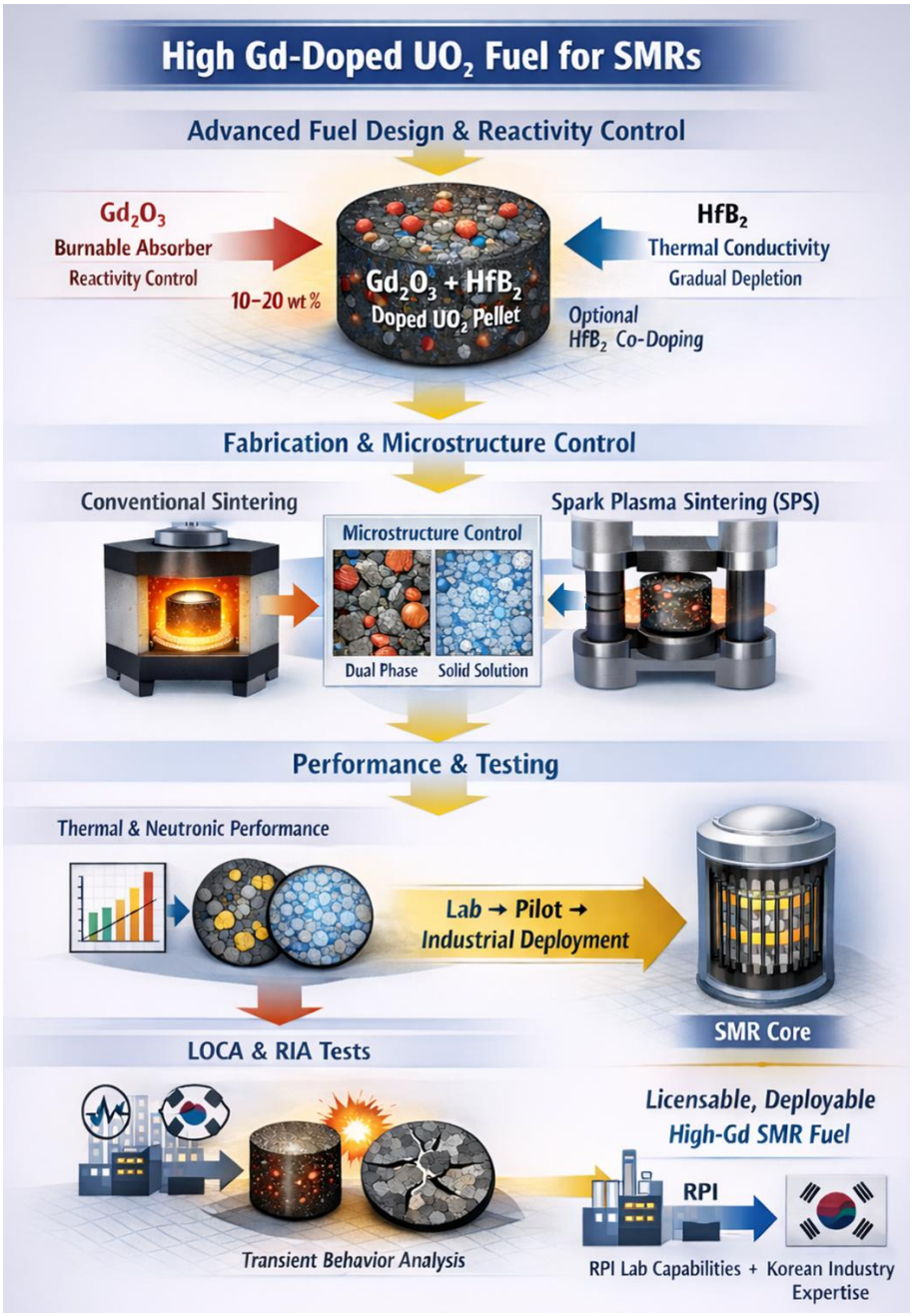
2. Technology Overview

Title of Proposed Technology	**Development of High Gadolinium-Doped Nuclear Fuels for Small Modular Reactors (SMRs)**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (x), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Nuclear Power	*Nuclear Fuel and Component Material*	*Nuclear Power*	*Next-Generation Nuclear Power*
Concept of Proposed Technology	The proposed technology focuses on the development of high Gd-doped UO_2 fuel to address reactivity control, thermal performance, and operational simplicity in Small Modular Reactors (SMRs) using high-assay low-enriched uranium (HALEU) [1-4]. The fuel integrates high Gd_2O_3 loading (up to ~20 wt%), with optional co-doping of hafnium diboride (HfB_2), to provide robust, in-fuel burnable absorber functionality, thereby reducing or eliminating dependence on soluble boron systems. By combining the strong early reactivity suppression of Gd_2O_3 with the gradual depletion and high thermal conductivity of optional HfB_2, this approach mitigates the limitations of conventional high-Gd fuels, including reduced thermal conductivity [5,6] and localized power peaking. Fuels will be fabricated via conventional sintering and spark plasma sintering (SPS), allowing precise control of dopant distribution, microstructure, and phase stability to optimize both neutronic and thermal performance (see Figure 1 for the schematic overview of the proposed concept). The embedded burnable absorbers enable simplified SMR operation, higher burnup, and increased uranium enrichment while maintaining mechanical integrity. Fuel behavior under steady-state and transient conditions, including LOCA and RIA scenarios, will be experimentally characterized using controlled high heating rates, providing predictive insight into cracking, thermomechanical response, and safety margins.			

	The outcome is a scalable, licensable fuel technology that expands SMR core flexibility, enhances safety margins, and aligns with industry priorities for simplified operation, high availability, and commercialization, establishing a clear pathway for industry collaboration from feasibility through deployment.  ***Figure 1. Conceptual schematic of high Gd-doped UO_2 fuel development for SMRs,*** *illustrating dopant integration, fabrication pathways (conventional sintering and SPS), microstructural control, thermal-neutronic performance, transient testing, and deployment through industrial collaboration.*
Rationale for Joint R&D Project	RPI group has extensive experience in fuel design, manufacturing and performance evaluations on a wide range of fuels from oxide, nitride, silicide to metal fuels, and is the key partner to DOE National Laboratories and leading fuel vendors such as Westinghouse, as demonstrated by extensive publications on nuclear fuels [7-14]. This project is also based on the preliminary results of the PI on the demonstration of 10 wt% Gd_2O_3 into UO_2 fuels (see Figure 2). Developing higher Gd-doped UO_2 fuel for SMRs requires integrating advanced fuel fabrication, high

	dopant incorporation, systematic microstructure/phase and thermo-physical properties characterization and transient performance testing with realistic SMR core designs and manufacturing expertise. Collaboration with Korean industry is essential because they can provide SMR-specific core data, neutronic simulation, and regulatory guidance, while our team contributes unique laboratory-scale capabilities in high-Gd fuel fabrication, thermal-mechanical characterization, and transient testing. The Korean industry can also lead the transition from lab-scale research feasibility demonstration to scale-up production. This joint effort ensures the fuel design is both innovative and deployable, bridging the gap between research and industrial implementation.
Key Words	1. **High-Burnup Fuel –** Designed for extended cycle operation in compact SMRs. 2. **HALEU Fuel –** up to 20 wt% low-enriched uranium for higher reactivity and power density. 3. **Burnable Poison (Gd_2O_3 up to 20 wt%) –** Controls excess reactivity and power peaking without soluble boron. 4. **Compact SMRs –** Small-core design requiring advanced reactivity management. 5. **Thermal-Mechanical & Transient Performance –** Characterization under steady-state and power transient conditions to evaluate fuel performance and ensure safe operation.

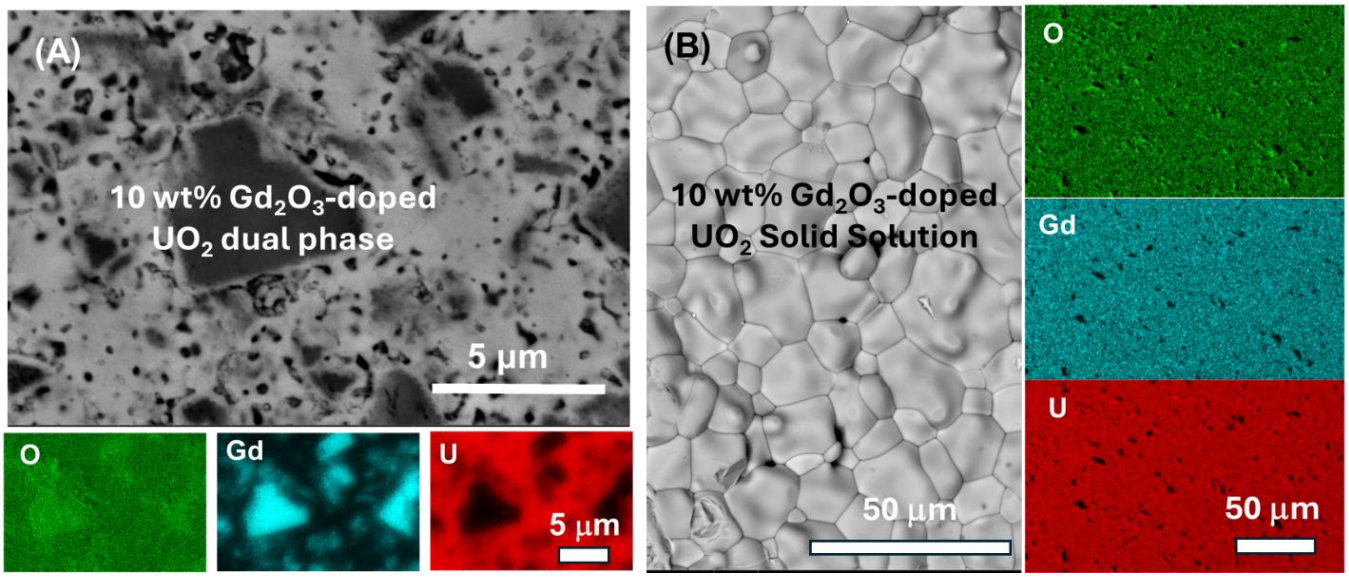


Figure 2. Preliminary results from the PI demonstrate the successful fabrication of UO_2 fuels doped with 10 wt% Gd_2O_3 using both conventional sintering and spark plasma sintering (SPS), enabling controlling the phase behavior either in dual phase composites (A) or a single-phase solid solution (B) ***(unpublished results)***. Further research is needed to assess the viability of higher Gd-doping, up to 20 wt%, including the potential for co-doping, and how these impact fuel thermophysical properties and transient performance, forming the basis of the proposed research.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2-3)	End of the Project	Level (4-5)

R&D Objectives

1. **High Gd-Doped UO_2 Fuel Fabrication**: Demonstration of the fabrication of UO_2 pellets with up to 20 wt% Gd_2O_3 with optional HfB_2 co-doping using conventional sintering and SPS, achieving uniform microstructure, high density, and controlled grain size suitable for SMR cores.
2. **Optimized Burnable Absorber Design:** Engineer Gadolinium burnable poison distributions with optional HfB_2 co-doping to control excess reactivity in compact SMR cores without relying on soluble boron for extended burnup and cycle length.
3. **Thermal-Mechanical Characterization:** quantify thermal conductivity and mechanical strength of high-Gd fuels as key metrics for the evaluation of fuel performance
4. **Transient Testing and Safety Assessments:** performing transient testing of high-Gd fuels under controlled rapid heating (up to several hundred °C/s) to simulate LOCA and RIA accident scenarios, enabling assessment of fuel performance, cracking, and fragmentation under prototypical accident conditions.
5. **Integrated Fuel Design & Scale-Up Roadmap**: Combine fabrication, characterization, and transient data to produce deployable fuel designs and a roadmap for pilot-scale production, including collaboration with Korean industry for SMR-specific adaptation and quality assurance.

R&D Target Specifications

Target Performance Specifications

Parameter	Target Value / Range	Notes
Gd_2O_3 Loading	Up to 20 wt%	High burnable absorber content
Pellet Density	>95% theoretical density	Both conventional sintering or SPS
Thermal Conductivity	Comparable with UO_2 without significant degradation	Measured at operating temperatures
Transient Response	Evaluating fracture/fragmentation behavior	Heating rates 1–500 °C/s simulating LOCA/RIA

R&D Plan

Phase 1 (Months 0-6): Feasibility Exploration and Full-scale R&D Plan

Objective: Develop a full research plan and assess the feasibility and technical space for developing high Gd-doped UO_2 fuels for compact SMRs.

- Establishment of full-scale R&D plans, experiments and demonstration of plans, market analysis, and IP discussion
- Evaluate the technical feasibility of high Gd-doped UO_2 fuel for compact SMRs focusing on assessing whether the combination of high Gd loading or with HfB_2, HALEU enrichment, and SMR core requirements is achievable within material, fabrication, and safety constraints.

Phase 2a: Proof-of-Concept Demonstration (Months 7–12)

Objective: Demonstrating the feasibility of developing high Gd-doped UO2 fuel concepts for SMRs and assessing neutronic performance.

(a) Preliminary Neutronics Assessment and Burnable Absorber Evaluation (by Industry collaborator)

- Perform initial neutronics calculations (e.g., simplified MCNP or SCALE models) to estimate feasible Gd content (10–20 wt%) and isotopic composition ($^{155}Gd/^{157}Gd$) with optical HfB_2

co-doping.

- Assess reactivity control, power peaking, and preliminary cycle length for different Gd loading scenarios in a representative SMR core.
- Evaluate whether soluble-boron-free operation is achievable with planned Gd concentrations and HALEU enrichment levels.
- Deliverable: Preliminary dopant and isotopic recommendation report with estimated impact on reactivity and core safety.
- Significance: Determines if high-Gd fuels with optional HfB_2 co-doping can meet SMR neutronics requirements, informing the decision to proceed to fabrication and testing.

(b) **Small-Scale Fuel Fabrication Trial (by RPI)**

- Fabricate laboratory-scale UO_2 pellets incorporating selected Gd levels using both conventional sintering or SPS methods.
- Evaluate densification, microstructural uniformity, and dopant distribution, and identify potential challenges in achieving target density (>95% TD).
- Conduct SEM/TEM analysis to observe grain size, porosity, and Gd clustering.
- Deliverable: Feasibility report on fabrication and microstructural integrity.
- Significance: Confirms whether high Gd loadings can be incorporated without unacceptable microstructural defects or inhomogeneities.

Phase 2b. Full Concept Development (Months 13-30)

Objective: Develop, optimize, and characterize high Gd-doped UO_2 fuels for compact SMRs, including fabrication, thermal-mechanical performance, transient behavior, and a roadmap for industrial scale-up.

(c) Neutronics and Burnable Absorber Optimization (Months 13–20) (led by Korean industry)

- Conduct high-fidelity neutronics simulations using MCNP to evaluate reactivity, power peaking, cycle length, and safety margins.
- Optimize Gd content and isotopic composition with optional HfB_2 co-doping, including spatial distribution within fuel pellets, to maintain soluble-boron-free operation.
- Examine the interaction between HALEU enrichment levels and Gd dopant to ensure adequate excess reactivity control over the full fuel cycle.
- Deliverable: Comprehensive report with optimal Gd dopant levels, isotopic composition, and expected SMR core performance metrics.
- **Significance:** Ensures that fuel design aligns with reactor safety and operational requirements.

(d) Fuel Fabrication and Steady-State Characterization (by RPI) (Months 13–30)

- Fabricate UO_2 pellets with 10–20 wt% Gd_2O_3 using optimized conventional sintering and SPS protocols.
- Powder processing: Ensure homogeneous mixing of UO_2 and Gd_2O_3 powders or solid solution, control particle size, and optimize pressing.
- Sintering optimization: Determine temperature, pressure, and dwell time to achieve >95% TD with controlled microstructure.
- Systematically investigating how co-doping of HfB_2 impacts the sinterability and microstructure control.
- Microstructural characterization: SEM/TEM imaging for grain size, porosity, dopant distribution, and potential clustering.
- Thermal conductivity measurements: investigating how different Gd_2O_3 doping with the optical HfB_2 co-doping impact the thermal conductivity of the fabricated fuels to achieve the goal matching with UO_2 fuels.
- Deliverable: Fabrication protocol and material property database linking Gd content,

processing conditions, and pellet performance.

- **Significance:** Provides validated fabrication procedures and establishes steady-state performance benchmarks for high-Gd SMR fuel.

Phase 2c – Transient Behavior and Safety Testing by RPI (Months 18–36)

- Conduct controlled rapid heating experiments simulating LOCA and RIA scenarios, with heating rates from 1–500 °C/s.
- Characterizing cracking, pellet deformation, and failure modes.
- Evaluate the impact of Gd content and microstructure on transient behavior, identifying safe operating thresholds (e.g., threshold heating rates and terminal temperature causing fracture).
- Deliverable: Transient performance dataset to validate predictive models for SMR safety analysis.
- Significance: Provides critical safety-relevant data to ensure high-Gd fuels can withstand accident scenarios without catastrophic failure.

Phase 2d – Integrated Fuel Design and Scale-Up Roadmap by RPI and Industry (Months 31–36)

- Integrate fabrication, steady-state characterization, and transient performance results to finalize high-Gd UO_2 fuel designs.
- Develop a pilot-scale fabrication plan, including quality assurance and process control measures for industrial production.
- Collaborate with Korean SMEs and industrial partners for potential technology transfer, demonstration, and SMR-specific adaptation.
- Address regulatory and licensing requirements to support future deployment of soluble-boron-free high-Gd fuel.
- Deliverable: Comprehensive final report including fuel design specifications, scale-up roadmap, and industrial collaboration plan.
- Significance: Bridges laboratory-scale R&D to industrial deployment, ensuring the technology is feasible, safe, and scalable.

Summary of Key Milestones and Timeline

Month	Milestone	Phase
6	Full research Plan and feasibility assessment	Phase 1
12	Optimized Gd dopant distribution & neutronics report	Phase 2a
18	Initial fabrication and microstructural assessment	Phase 2b
20	Steady-state thermal property database complete	Phase 2b
36	Transient testing and validated FEA models	Phase 2c
36	Final integrated fuel design and pilot-scale roadmap	Phase 2d

4. Role & Responsibility (R&R)

Expected R&R of Research Institute	**1. Technical Contributions:** • Lead the design and optimization of high Gd-doped UO_2 fuels for compact SMRs, including burnable absorber distribution, and microstructure control. • Provide extensive expertise in nuclear materials and fuel behavior, including thermal-mechanical performance and transient behavior under accident scenarios. • Material Science Expertise: Utilize in-house knowledge of ceramic and nuclear fuel materials to develop fabrication protocols, characterize material microstructure, and ensure compatibility with SMR operating conditions. **2. Research Scope:** • Fabrication of High-Gd UO_2 Fuel Pellets: Produce laboratory-scale pellets with 10–20 wt% Gd_2O_3 using both conventional sintering and spark plasma sintering (SPS) methods, ensuring high density and uniform dopant distribution. • Microstructure, phase and thermal property characterization: characterize the phase and microstructure of high Gd-doped fuels and measure thermal conductivity, mechanical strength. • Transient Behavior Assessment: Conduct controlled rapid heating experiments simulating LOCA and RIA scenarios, assessing transient thermal-mechanical response, cracking, and fragmentation. • Integrated Fuel Design and Scale-Up Roadmap: providing process parameters for potential pilot-scale fabrication. **3. Resources to be Provided:** • Laboratory infrastructure for powder processing, pellet fabrication, sintering, and thermal-mechanical properties characterization. These include sintering furnaces, SPS equipment, powder processing facilities, and characterization instruments (SEM, XRD), and a suite of thermomechanical properties (such as LFA, Dilatometer, simultaneous thermal analyzer, hardness testers and nano-indentation). • Ultrafast heating test apparatus with heating rate varying from a few ^{o}C/s up to several hundred ^{o}C/s for transient testing of fuel integrity and fracture/fragmentation behavior. • Technical Expertise: Experienced personnel in nuclear materials, fuel fabrication, thermal-mechanical characterization, and safety analysis. • Data management and reporting capabilities to document experimental results, simulations, and design decisions. **4. Development Tasks:** • Fabrication of high-Gd UO_2 pellets with controlled density and microstructure. • Comprehensive steady-state thermal-mechanical characterization. • Transient testing under LOCA/RIA conditions with FEA model validation. • Integrated fuel design and scale-up roadmap in collaboration with industry partners.
Expected R&R of Korean SME Partner	**1. Technical Contributions:** • Contribute practical industrial experience in reactor design and fuel optimization to guide the development of high Gd-doped fuels for SMRs. • Conduct neutronics and core simulations to predict reactivity, power peaking, burnup, and soluble-boron-free operation. • Performing fuel performance modeling (MCNP, FEA software) and reactivity analysis • Contribute practical industrial experience in fuel fabrication, processing, and pilot-scale production. • Implement industrial testing protocols to verify performance of high-Gd fuel under

	operational conditions. • Develop integrated fuel design and pilot-scale fabrication roadmap. **2. Development Tasks:** • Conduct neutronics and core simulations to predict reactivity, power peaking, burnup, and soluble-boron-free operation and guide the design of high Gd-doped fuels for SMRs. • Participate in transient testing or support experiments simulating reactor-relevant conditions if required. • Support the scale-up of fuel fabrication processes from laboratory to pilot scale. **3. Testing/Validation Activities:** • Implement industrial testing protocols to verify performance of high-Gd fuel under operational conditions. • Support the scale-up of fuel fabrication processes from laboratory to pilot scale. • Validate process scalability and quality control metrics for potential commercial deployment. **4. Resource Commitments:** • Allocate technical personnel for reactor and fuel design, process optimization, quality control, and validation activities. **5. Collaboration Efforts:** • Work closely with RPI to transfer laboratory findings to potential industrially viable fabrication methods. • Participate in joint data review meetings, design discussions, and technical workshops. • Leading the preparation of documentation for regulatory compliance and potential deployment.

5. Expected Outcome/Impact

Expected Outcomes and Impact	**1. Technological Achievements** • Development of High Gd-Doped UO_2 Fuel: Successful fabrication of laboratory-scale fuel pellets with up to 20 wt% Gd_2O_3 incorporated, with optional HfB2 co-doping, using conventional sintering and SPS, demonstrating high density (>95% TD) and uniform microstructure. • Comprehensive Fuel Characterization: Establish a robust dataset of steady-state thermal-mechanical properties, including thermal conductivity and detailed microstructural and microchemical analysis. • Transient Performance Insights: Demonstrate the behavior of high-Gd fuels under LOCA and RIA conditions using ultrafast controlled heating experiments, providing insights into safety margins and fuel resilience. • Integrated Fuel Design: Develop optimized fuel designs compatible with compact SMR cores, informed by Korean SME-led neutronics simulations and RPI experimental data. **2. Commercialization Potential** • Fuel Technology Transfer: Provide a pilot-scale fabrication roadmap for industrial deployment, enabling Korean SMEs and other nuclear fuel manufacturers to produce high-performance, high-burnup fuels for compact SMRs. • Soluble-Boron-Free Operation: Enable compact SMR operation with enhanced reactivity control using Gd burnable absorbers, reducing operational complexity and potentially lowering costs. **3. Broader Industrial and Economic Impact** • Support for SMR Deployment: Contribute to advancing compact SMR technologies with safer, high-burnup fuels, aligning with global trends in nuclear energy innovation. • Strengthened Industry-Academia Collaboration: Foster a strategic partnership between RPI and Korean SMEs, creating a model for international collaborative R&D in nuclear fuel development. • Economic Benefits: Promote domestic and international manufacturing capabilities, enabling competitive fuel production, technology licensing, and potential export opportunities. • Knowledge and Capability Building: Establish advanced experimental and simulation capabilities in fuel performance testing, contributing to workforce development and long-term scientific expertise in nuclear materials. **4. Deliverables and Measurable Outcomes** • High-density, high-Gd UO_2 fuel pellets characterized for thermal and mechanical properties. • Characterize their transient behavior and performance under accident scenarios. • Optimized fuel design is compatible with compact SMR operation. • Technical reports, publications, and potential patents related to high-Gd SMR fuels.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization